

Price Effects in Online Hotel Search

Causal Estimation and a Sequential Search Model

Anastasiia Rekovets

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Abstract

This paper estimates the causal effect of hotel prices on click-through and booking probability using the Expedia *Personalizing Hotel Searches* dataset: 9.9 million impressions across 399,344 search sessions. Identification addresses two sources of endogeneity – unobserved hotel quality and algorithmically determined display position – and leans on the subsample in which the ranking algorithm is switched off and hotels appear in random order. Eight specifications separate the sample restriction from the estimator change: pooled OLS, OLS on random-order sessions, within-session fixed effects, IV/2SLS and double/debiased machine learning. The preferred within-session estimate implies that a 10% price increase lowers click probability by roughly 0.28 percentage points, about 6% of the baseline click rate.

The IV specification is reported as a diagnostic rather than a preferred estimate. Its first stage is strong, but the correction runs in the opposite direction to what quality-driven endogeneity predicts, which is itself evidence against the exclusion restriction. A sequential search model (Ursu, Seiler and Honka, 2024), estimated by simulated maximum likelihood, then separates consumer preferences from position-dependent search costs, which rise about 4.7% per position down the page.

Reproducibility. Every number, table and figure in this paper is taken directly from the saved outputs of the accompanying notebook, and this is enforced by an automated check. Table and figure sources are named in the notes to each exhibit.

AI tool usage. Claude was used as a coding assistant for syntax and debugging, and for a review pass over the analysis and write-up.

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1 Data and Summary Statistics

The analysis uses the Expedia *Personalizing Hotel Searches* dataset (Expedia, Inc., 2013), processed out-of-core with `polars`. The training file contains 9,917,530 hotel-search impressions across 399,344 unique search sessions and 136,886 unique properties, covering 1 November 2012 to 30 June 2013, with an average of 24.8 hotels listed per search. The outcomes are `click_bool` (4.47% overall, 443,672 clicks) and `booking_bool` (2.79%, 276,593 bookings). No booking occurs without a click, and $P(\text{book} \mid \text{click}) = 62.34\%$.

Assessment. The scale of the data and the real financial stakes make it well suited to studying price-demand relationships. Its central strength is the `random_bool = 1` subsample (2,939,652 rows, 29.6% of the data), in which hotels are displayed in random order. This provides strictly exogenous variation in display position and is the basis for the preferred specifications below.

Three limitations deserve emphasis. First, missingness is severe in exactly the variables that would be most valuable: `gross_bookings_usd` is 97.2% missing (precluding revenue-weighted analysis), the `visitor_hist_*` fields are 94.9% missing (individual price sensitivity is unobserved), and the competitor price fields are 55–98% missing, reflecting structural absence – the hotel is not listed on that OTA – rather than random non-response. Second, the platform sample is selected: these are Expedia searches, not the travel market. Third, and most important for interpretation, the random-order subsample behaves oddly. Its click rate is *higher* than in ranked sessions (4.66% vs 4.40%), while its booking rate is roughly seven times *lower* (0.54% vs 3.74%). A pure position effect would not produce a seven-fold booking gap alongside a higher click rate. The more cautious reading is that random-order clicks partly reflect browsing curiosity rather than purchase intent, and that random-order sessions may not be a clean random draw from the same population. Estimates identified off this subsample (M3r, M4, M6 below) should be read with that in mind, particularly for bookings.

Table 1: Summary Statistics for Continuous Variables

Variable	N	Mean	Std	Min	P25	P50	P75	Max
<code>price_usd</code>	9,917,530	241.78	14,341.81	0.00	85.00	122.07	185.00	19,726,328
<code>log_price_usd</code>	9,917,476	4.85	0.62	−4.61	4.44	4.80	5.22	16.80
<code>prop_log_historical_price</code>	9,917,530	4.32	1.84	0.00	4.44	4.91	5.31	6.21
<code>prop_starrating</code>	9,917,530	3.18	1.05	0.00	3.00	3.00	4.00	5.00
<code>prop_review_score</code>	9,902,900	3.78	1.05	0.00	3.50	4.00	4.50	5.00
<code>prop_location_score1</code>	9,917,530	2.88	1.53	0.00	1.79	2.77	4.04	6.98
<code>srch_length_of_stay</code>	9,917,530	2.39	2.07	1.00	1.00	2.00	3.00	59.00
<code>srch_booking_window</code>	9,917,530	37.62	52.11	0.00	4.00	17.00	49.00	498.00
<code>position</code>	9,917,530	16.87	10.43	1.00	8.00	16.00	26.00	40.00

Notes: Unit of observation is a hotel-search-session impression. `price_usd` is extremely right-skewed (54 rows are exactly zero and 5,707 rows exceed \$5,000; both are almost certainly data errors), which is why all models use `log_price_usd`. The 54 zero-price rows are set to null and excluded from every regression. A further 1,430,427 rows have `prop_log_historical_price` = 0, a missing-value sentinel implying a \$1 historical rate; these are dropped where that variable is used as a regressor or instrument. Source: notebook cell 7.

Quality signals also shift baseline rates substantially. Click rates by star rating are non-monotone and peak at four stars (5.39%), above both three-star (4.38%) and five-star (4.59%) properties; unrated and one-star properties sit lowest (2.90% and 2.80%). The largest single gap in the data comes from promotions: properties with an active promotion are clicked 6.04% of the

Table 2: Click and Booking Rates by Log-Price Decile

Decile	<i>N</i>	Median price (\$)	Click rate (%)	Booking rate (%)
1 (cheapest)	995,902	51.00	4.37	2.81
2	987,605	70.00	4.64	3.13
3	991,776	85.00	4.94	3.36
4	993,419	99.00	4.98	3.28
5	990,038	114.76	4.89	3.19
6	1,002,811	131.06	4.83	3.07
7	1,049,587	155.00	4.58	2.79
8	928,649	186.77	4.37	2.56
9	985,963	234.11	4.05	2.26
10 (most exp.)	991,726	356.00	3.07	1.41

Notes: Source: notebook cell 12. Demand is *not* monotonically decreasing in price. Both click and booking rates rise from decile 1 to a peak at decile 3–4 (booking 3.36% at a median of \$85) before declining steadily; the fall is steepest above decile 8. The upturn at the bottom is economically meaningful rather than noise: the very cheapest listings attract fewer clicks than mid-priced ones, consistent with low price acting as a negative quality signal. A purely monotone reading of the raw decile gradient would miss this.

time versus 4.04% without, and booked 3.92% versus 2.48%.

2 Visualising Price Effects

Visual analysis isolates the endogeneity of display position. Figure 1a shows click and booking rates across raw price deciles pooled over all sessions. Figure 1b restricts to random-order sessions, where the gradient flattens substantially. The difference is direct evidence that Expedia’s ranking algorithm confounds the price effect by systematically placing cheaper hotels higher on the page.

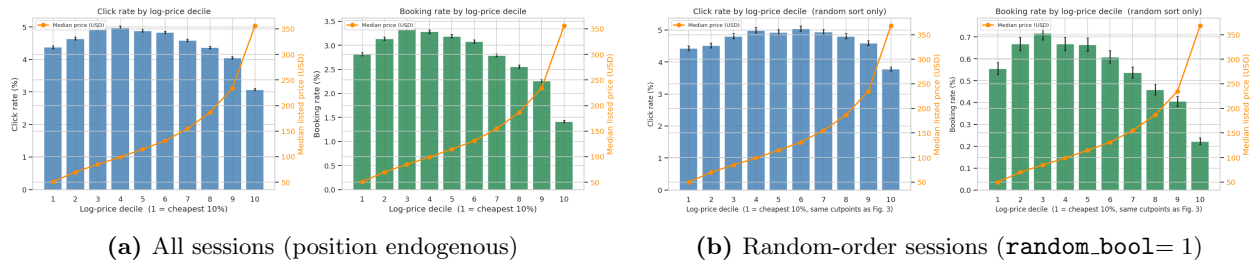


Figure 1: Click and booking rates by log-price decile. The flattening in (b) isolates the position confound. Source: notebook cells 12–13.

The sharpest descriptive evidence comes from comparing a hotel to the alternatives shown in the *same* session, which absorbs all session-level confounders. Figure 2 plots outcomes against within-search price rank in random-order sessions: the cheapest hotel in a session is clicked 8.67% of the time, falling to 6.22% at rank 5 and 4.62% at rank 10 – a $1.39\times$ ratio between ranks 1 and 5, with a corresponding $1.26\times$ ratio for bookings.

Figure 3 shows that booked hotels concentrate at lower prices and that price sensitivity persists strictly within star-rating tiers, so the gradient is not simply quality sorting. A useful diagnostic sits in the same cell: among ranked sessions, clicked hotels are on average \$12.0 cheaper than unclicked ones, but in random-order sessions the gap narrows to \$5.6 – again indicating that a large share of



Figure 2: Outcomes by within-search price rank (`random.bool=1`). Source: notebook cell 14.

the raw price–click association is a display artefact.

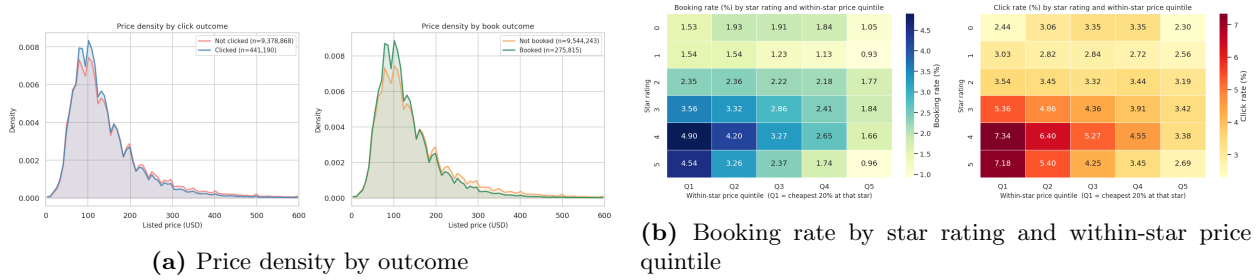


Figure 3: Price distributions and quality–price interactions. Source: notebook cells 15–16.

3 The Causal Effect of Price on Demand

To separate the causal price effect from unobserved quality and display position, I estimate a series of linear probability models,

$$y_{ij} = \alpha + \beta \log p_{ij} + \gamma' \mathbf{X}_{ij} + \delta \text{position}_{ij} + \varepsilon_{ij}, \quad (1)$$

where β is a semi-elasticity: a 1% price increase changes the outcome probability by $\beta/100$ percentage points. Eight specifications are estimated:

- **M1–M3 (OLS).** Progressively add hotel quality controls and display position, on a 2M-row sample.
- **M3r.** M3 re-estimated on random-order sessions only. Comparing M3 to M3r isolates the *sample* restriction.
- **M3.IV.** M3 re-estimated on the exact 500k rows used for M5, so that the M3.IV→M5 comparison is a clean OLS-versus-IV contrast unconfounded by sample composition.
- **M4 (within-session FE).** Restricted to `random.bool=1` and demeaned within `srch_id` by the Frisch–Waugh–Lovell theorem, absorbing every search-level unobservable. SEs clustered on session.
- **M5 (IV/2SLS).** Instruments price with `prop_log_historical_price`.

- **M6 (DDML).** Partially linear regression with histogram gradient-boosted nuisance functions and cross-fitting, relaxing linearity in controls.

Table 3: Price Coefficient Estimates Across Specifications

Model	Estimator	N	Click coef.	Booking coef.
M1: Bivariate OLS	OLS, HC1	2,000,000	-0.0063*** (0.0002)	-0.0071*** (0.0002)
M2: +Controls	OLS, HC1	2,000,000	-0.0143*** (0.0003)	-0.0116*** (0.0002)
M3: +Position	OLS, HC1	2,000,000	-0.0148*** (0.0003)	-0.0120*** (0.0002)
M3r: +Pos. (random)	OLS, HC1 (random)	2,933,760	-0.0114*** (0.0002)	-0.0025*** (0.0001)
M3.IV: +Pos. (IV sample)	OLS, HC1 (IV sample)	500,000	-0.0145*** (0.0005)	-0.0118*** (0.0004)
M4: Within-session FE	FE, clustered SE	2,933,760	-0.0298*** (0.0004)	-0.0049*** (0.0001)
M5: IV / 2SLS	IV2SLS, robust SE	500,000	-0.0074*** (0.0008)	-0.0092*** (0.0006)
M6: DDML (HGBR)	DDML PLR, cross-fit	150,000	-0.0134*** (0.0011)	-0.0024*** (0.0003)

Notes: Source: notebook cell 25. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All coefficients are semi-elasticities on log price, including M4’s within-session demeaned log price, so magnitudes are directly comparable across rows. Every coefficient in the table, M6’s booking estimate included, is significant at the 1% level. The preferred specification is M4.

Interpretation and magnitudes. Price reduces demand in every specification. Taking M4 as the preferred estimate, a 10% price increase ($\Delta \log p = 0.0953$) lowers click probability by $0.0298 \times 0.0953 = 0.28$ percentage points. Against the 4.66% baseline click rate in random-order sessions, that is a decline of about 6% in relative terms. The corresponding booking effect is $0.0049 \times 0.0953 = 0.047$ percentage points, roughly 8.6% of the 0.54% random-order booking base rate. The much smaller absolute booking coefficient therefore does not indicate weaker price sensitivity for bookings; it reflects the very low booking base rate in this subsample, and the two relative magnitudes are broadly comparable.

The decomposition built into the table is worth reading directly. Moving from M3 (-0.0148) to M3r (-0.0114) is the pure *sample* effect of restricting to random-order sessions; moving from M3r to M4 (-0.0298) is the pure *estimator* effect of within-session demeaning. The within-session estimate is roughly twice the pooled OLS estimate, which is what we expect if session-level unobservables (destination, dates, party composition, and the overall price level of the choice set) mask price sensitivity in the pooled data.

The IV specification is a diagnostic, not a preferred estimate. The first stage is strong on conventional criteria: the instrument enters with a coefficient of 0.8901 (SE 0.0017), the partial F -statistic is 261,009.8, and the Shea partial R^2 is 0.43078. Weak-instrument bias is therefore not the binding concern here, and it would be wrong to dismiss M5 on those grounds.

The binding concern is the exclusion restriction, and the estimates themselves supply evidence against it. The instrument is the same hotel’s own historical Expedia rate, which plausibly proxies

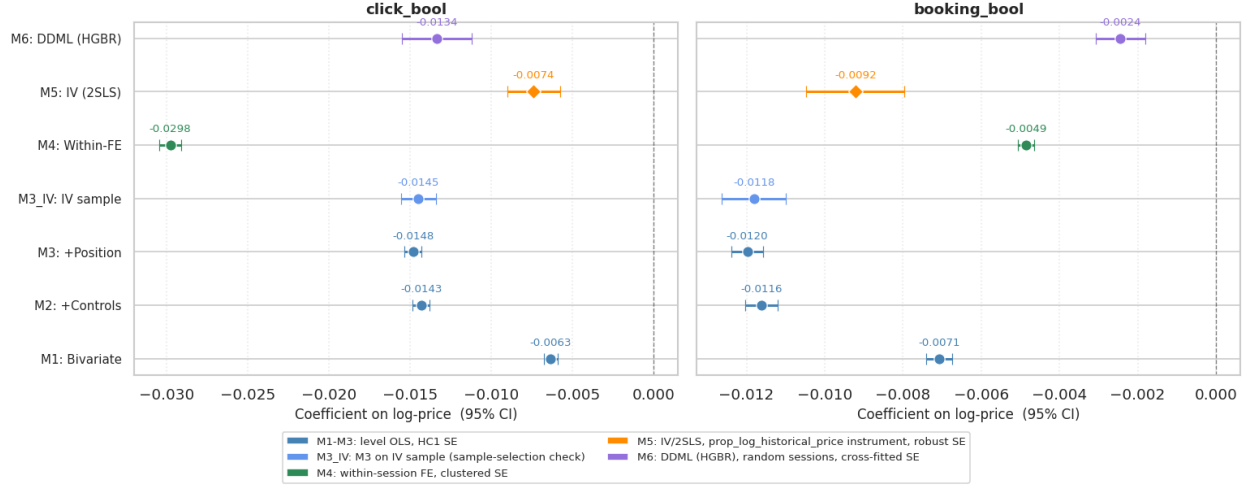


Figure 4: Price coefficient semi-elasticities across specifications (95% CI). Source: notebook cell 24.

for persistent unobserved quality and amenities that enter demand directly. The near-unit pass-through from historical to current price is exactly what a persistent, quality-driven price level would generate, so the strong first stage is as much a warning as a reassurance.

The direction of the correction makes this concrete. On the identical 500,000 rows, OLS gives -0.0145 (click) and -0.0118 (booking), while 2SLS gives -0.0074 and -0.0092 : the IV estimates are *less* negative than OLS. The standard quality-endogeneity story predicts the opposite. If expensive hotels are systematically better in ways we do not observe, and quality raises demand, then OLS is biased toward zero and a valid instrument should push the estimate *more* negative. Obtaining a correction in the wrong direction is the signature of an instrument that carries the very quality signal it was meant to purge. The Wu–Hausman tests reject exogeneity (click statistic 124.27, booking 26.29, both $p < 0.0001$), but at $N = 500,000$ that test rejects on economically trivial differences and does not rescue the design. M5 is reported here for completeness and as evidence about the instrument; it is not used as a causal estimate.

4 A Sequential Search Model

Following Ursu, Seiler and Honka (2024), consumer utility for hotel j is

$$u_{ij} = \delta_{ij} + \mu_{ij} + \varepsilon_{ij}, \quad \delta_{ij} = \mathbf{X}_{ij}'\boldsymbol{\beta}, \quad (2)$$

where μ_{ij} and ε_{ij} are standard normal pre- and post-search shocks. The reservation utility equates the marginal gain from searching with its cost, $z_{ij} = \delta_{ij} + \mu_{ij} + m(c_{ij})$, and search costs are position-dependent, $c_{ij} = \exp(\gamma_0 + \gamma_1 \cdot \text{position}_{ij})$.

The model is estimated by simulated maximum likelihood on 2,000 random-order sessions with $D = 300$ draws, drawn from the 38,937 sessions with at least one click and between 2 and 20 listings. Of the estimation sample, 276 sessions (13.8%) end in a booking and 1,724 (86.2%) in no purchase. Standard errors come from a subsampled bootstrap (Politis–Romano) with $B = 100$ replicates of $m = 500$ sessions; 2 replicates failed to converge.

Table 4: Sequential Search Model Estimates

Parameter	Estimate	Approx. SE	<i>t</i> -stat	95% CI
β_{price} (log price)	-0.6454	0.0098	-66.17	[-0.665, -0.626]
β_{star} (star rating)	+0.0614	0.0006	96.16	[+0.060, +0.063]
β_{review} (review score)	+0.1254	0.0019	66.06	[+0.122, +0.129]
β_{location} (location score 1)	-0.0560	0.0006	-95.17	[-0.057, -0.055]
β_{brand} (brand hotel)	+0.0216	0.0002	127.55	[+0.021, +0.022]
β_{promo} (promotion flag)	+0.0300	0.0002	136.07	[+0.030, +0.030]
β_0 (base hotel intercept)	-0.6023	0.0085	-71.14	[-0.619, -0.586]
γ_0 (log baseline search cost)	-1.2547	0.0222	-56.39	[-1.298, -1.211]
γ_1 (position effect on cost)	+0.0457	0.0007	62.42	[+0.044, +0.047]

Notes: Source: notebook cell 32. $N = 2,000$ sessions, $D = 300$ draws, subsampled bootstrap $B = 100$. $\sigma_\mu = \sigma_\varepsilon = 1$ is a normalisation, so coefficient magnitudes are interpretable only relative to one another. The reported standard errors should be treated as indicative rather than exact; see the discussion of their plausibility below.

Results. The structural parameters separate preferences from search frictions. Expected utility falls with price ($\hat{\beta}_{\text{price}} = -0.645$), and star rating, review score, brand and promotion all enter positively. Since the scale is normalised, magnitudes are meaningful only in relative terms: one additional star is worth about 0.095 log-price units and a one-unit increase in review score about 0.194 log-price units. The negative coefficient on `prop_location_score1` is contrary to expectation and is best read as that variable proxying for something else once price and the other quality measures are conditioned on, rather than as consumers disliking good locations.

Search costs rise with depth on the page ($\hat{\gamma}_1 = +0.046 > 0$), implying about 4.7% higher cost per position and making position 20 roughly $2.38\times$ as costly to reach as position 1. In levels, the implied cost rises from 0.299 at position 1 to 0.712 at position 20. Forward-simulating the search process at $\hat{\theta}$ reproduces the observed behaviour reasonably well: 1.03 model clicks per session against 1.10 observed, with a click gradient of 13.4%, 8.7%, 5.0%, 3.1% across position buckets 1–5, 6–10, 11–15 and 16–20, against observed values of 15.8%, 8.4%, 4.5% and 2.7%. Fit deteriorates in the 21–40 bucket (2.0% model versus 3.8% observed).

Limitations. Four caveats apply, and the first two are substantial.

First, the reported standard errors are implausibly small. *t*-statistics between 62 and 136 on a 2,000-session simulated likelihood with $D = 300$ are not credible; they almost certainly reflect the subsampling rescaling in the bootstrap rather than genuine precision. The point estimates should be taken as the informative output of this exercise and the standard errors treated as a known open problem, to be replaced by a properly rescaled subsampling variance or a numerical Hessian before any inference is drawn. I flag this rather than report inference I do not believe.

Second, 7.6% of sessions sit at the 10^{-10} likelihood floor, meaning the log-likelihood is partly driven by the floor rather than by the model; D should be raised until that share is negligible.

Third, identification of the search-order rule is weak: sessions average only 1.11 clicks, so most of the information comes from stopping and choice rather than from the order in which alternatives are inspected. Relatedly, the data contain no intra-session timestamps, so the likelihood assumes consumers search in top-to-bottom display order.

Fourth, $\sigma_\varepsilon = 1$ is a normalisation, so search costs cannot be monetised, and $D = 300$ with $N = 2,000$ was chosen for speed; production work would use $D \geq 500$ and the full session set.

Finally, the structural and reduced-form estimates are not directly comparable in magnitude. β_{price} is on a utility scale while the reduced-form coefficients are probability-scale semi-elasticities; they agree in sign, and the value the structural model adds is separating the preference channel from the search-friction channel that the reduced form bundles together.

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